

Multi-Temporal Mapping of Forest Disturbance and Recovery

Alluri Sitaramaraju District, Eastern Ghats (2018–2024)

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Project Statement and Scope

- Forest ecosystems in the Eastern Ghats are increasingly affected by human activities and climate variability.
- This project analyzes forest disturbance and recovery patterns using multi-temporal satellite data.
- Study focuses on the Alluri Sitaramaraju district between 2018 and 2024.
- Remote sensing techniques are used to detect changes in vegetation cover and fire risk.

Why This Research Matters

- Forest disturbances threaten biodiversity and ecosystem stability.
- Understanding disturbance patterns helps in better forest management.
- Satellite monitoring enables large-scale and continuous environmental assessment.
- Results can support conservation planning and climate resilience strategies.

Objectives of the Study

- Analyze forest disturbance patterns in Alluri Sitaramaraju district from 2018–2024
- Detect vegetation changes using multi-temporal satellite imagery
- Identify areas affected by forest fires and human activities
- Examine forest recovery trends after disturbance
- Support sustainable forest management using remote sensing insights

Study Area Context

- Alluri Sitharama Raju district located in the Eastern Ghats of Andhra Pradesh (17°–18°N)
- Highly forested region with about **64% forest cover**
- Forest types include dry and moist deciduous forests
- Region largely managed by tribal communities

Drivers of Forest Disturbance

- Human activities such as shifting cultivation, logging and grazing
- Road expansion and settlement growth
- Frequent forest fires during dry seasons
- Climate change causing rainfall variability and drought stress

- Landsat 8 and Landsat 9 satellite imagery
- Sentinel-2 multispectral imagery
- MODIS NDVI vegetation datasets
- CHIRPS rainfall datasets
- Government land cover datasets (APSAC, DSR reports)

Remote Sensing Approaches

- Multi-temporal satellite imagery used to monitor forest change
- Vegetation indices such as NDVI used to detect disturbance
- Time-series analysis helps detect long-term trends
- Platforms like Google Earth Engine enable large-scale analysis

Study Area

- Located in the **Eastern Ghats of Andhra Pradesh** (17° – 18° N)
- Forest area: **7,838 km²** (**64%** of district area)
- Dominant forests: **dry and moist deciduous**
- Terrain: **mountainous and rugged**
- Important biodiversity hotspot



- **Step 1 — Fire Date Determination:** MODIS FIRMS, NASA VIIRS and FSI calendars cross-validated to fix active fire season (Feb–Apr 2023).
- **Step 2 — Fuel Danger Index (FDI):** NDVI and NDMI derived from MODIS and Landsat 9 to quantify fuel availability and moisture.
- **Step 3 — Activity Danger Index (ADI):** Human accessibility and population density used to capture anthropogenic fire pressure.
- **Step 4 — Weather & Topographic Danger Indices (WDI / TDI):** ERA5-Land climate variables and SRTM terrain data.
- **Step 5 — dNBR (Burn Severity):** Pre- and post-fire Landsat 9 NBR differencing.
- **Step 6 — AHP Risk Modelling:** Eight-factor weighted linear combination to produce composite wildfire risk map.

Fire Date Determination

Objective: Identify the active fire season for ASR district to anchor pre- and post-fire analysis windows.

Data Sources & Approaches:

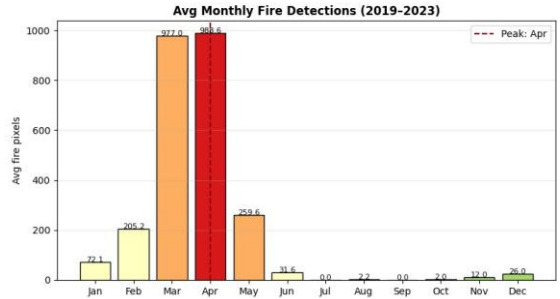
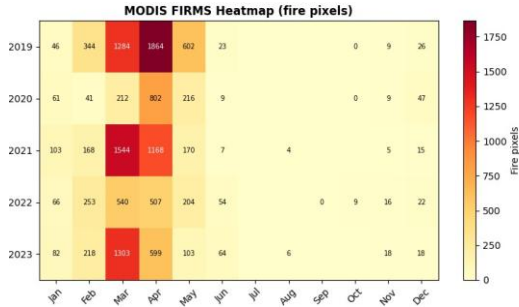
- **Approach A — MODIS FIRMS (GEE):** Monthly fire pixel counts (2019–2023); T21 brightness temperature > 300 K threshold.
- **Approach B — FSI Fire Calendars:** ISFR 2021 & 2023 confirm Feb–Apr as the peak fire-risk window for Andhra Pradesh.

Outcome: Both approaches converge on **Feb 15 – Apr 30, 2023** as the confirmed fire season; this window defines pre-fire and post-fire imagery selection for all subsequent indices.

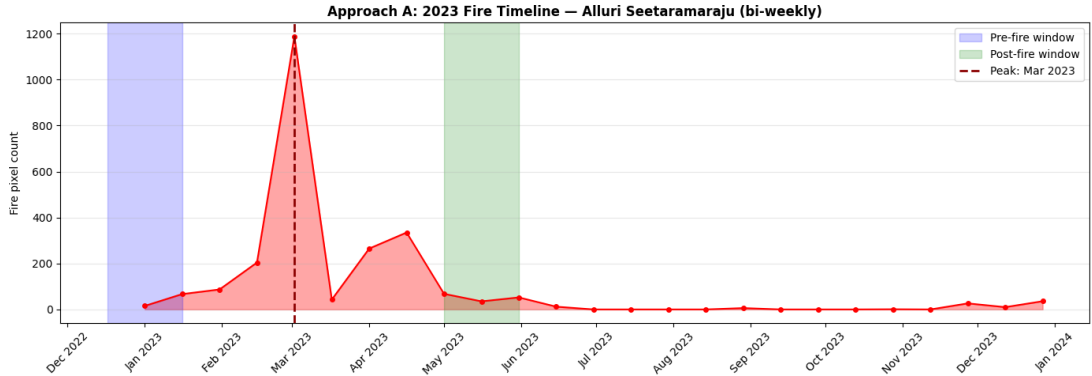
[FIRMS: **Fire Information for Resource Management System: Dev by NASA**]

Approach A — MODIS FIRMS (GEE)

Approach A: MODIS FIRMS — Alluri Seetaramaraju District

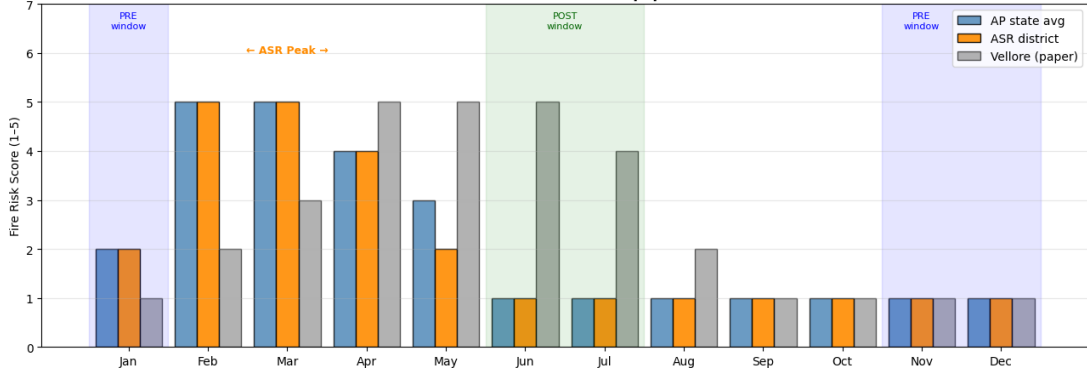


Approach A — MODIS FIRMS (GEE)



Approach B — FSI Fire Calendars

Approach C: FSI Fire Season Calendar
ASR vs AP State vs Vellore (paper site)



Formulae Used

NDVI (Normalized Difference Vegetation Index)

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

NDMI (Normalized Difference Moisture Index)

$$\text{NDMI} = (\text{NIR} - \text{SWIR}_1) / (\text{NIR} + \text{SWIR}_1)$$

FDI (Fuel Danger Index)

$$\text{FDI} = (\text{NDVI} \times T) / M \text{ or } (\text{NDVI} + \text{NDMI}) : [T: \text{Temp}, M: \text{Moisture}]$$

ADI (Activity Danger Index)

$$\text{ADI}(\text{road}) = 1 - (\text{Dmax} - \text{Dmin} / \text{Dr} - \text{Dmin})$$

ADI(Settle) is inversely proportional to distance to settlements

$$\text{ADI} = (\text{ADIr} + \text{ADIs}) / 2$$

WDI (Weather Danger Index)

$$\text{WDI} = (\text{Tcls} + \text{RHcls} + \text{WScls}) / 3$$

Term

Meaning

(T_{cls})

Temperature danger class

(RH_{cls})

Relative humidity danger class

(WS_{cls})

Wind speed danger class

Formulae Used (cont.)

TDI (Temperature Danger Index)

$$\text{TDI} = (T - T_{\min}) / (T_{\max} - T_{\min})$$

NBR (Normalized Burn Ratio)

$$\text{NBR} = (\text{NIR} - \text{SWIR}_2) / (\text{NIR} + \text{SWIR}_2)$$

dNBR (Differenced Normalized Burn Ratio)

$$\text{dNBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}$$

Data Source: MODIS MOD13Q1 **Spatial Resolution:** 250 m **Temporal Resolution:** 16-day composites

Study Periods:

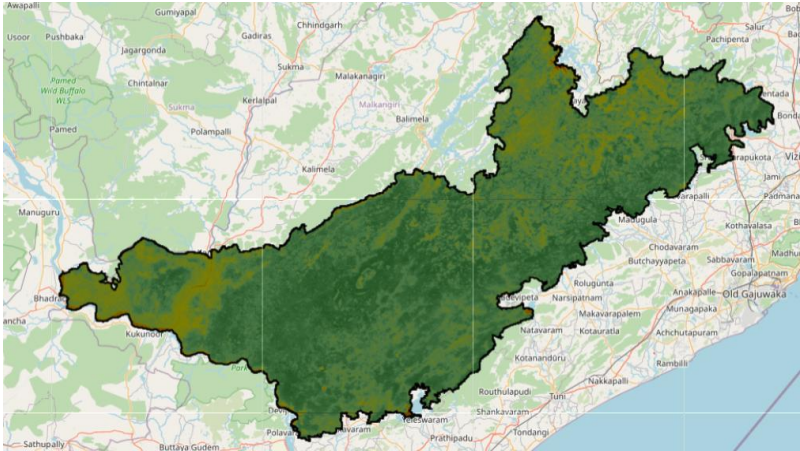
- Pre-fire: October 2022 – January 2023
- Post-fire: May – July 2023

Role in Fire Analysis:

- NDVI represents vegetation greenness and biomass density.
- Higher NDVI → dense vegetation → higher fuel availability under dry conditions.
- Lower NDVI indicates sparse vegetation or already-burned areas.

Processing:

- MODIS NDVI resampled from 250 m to 30 m via bilinear interpolation.
- Ensures spatial consistency when integrating with Landsat-derived layers in the composite fire risk model.



NDMI — Fuel Moisture Indicator

Data Source: Landsat 9 Surface Reflectance (B5) and SWIR-1 (B6)

Spatial Resolution: 30 m

Bands: NIR

Study Periods:

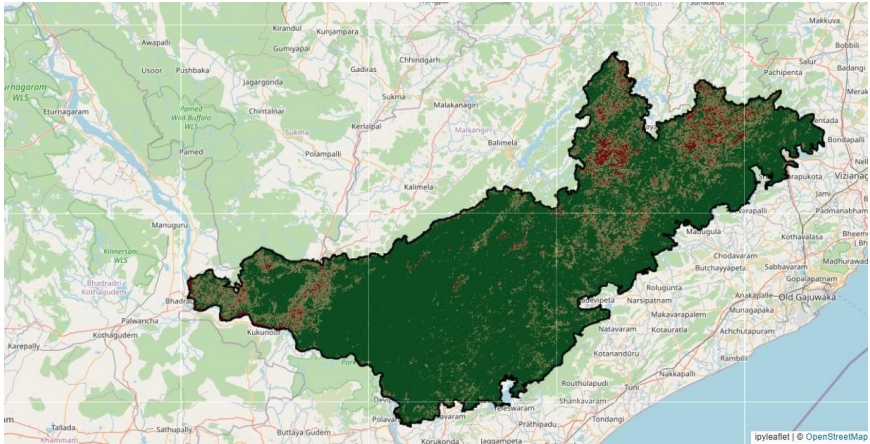
- Pre-fire: October 2022 – January 2023
- Post-fire: May – July 2023

Role in Fire Analysis:

- NDMI captures water content in vegetation canopy.
- Low NDMI → dry(red), moisture-stressed vegetation → elevated ignition risk.
- Complements NDVI by distinguishing green but dry fuel from genuinely moist vegetation.

Processing:

- Cloud and shadow masking applied using QA_PIXEL band.
- NDMI composited over the fire-season window and clipped to ASR district boundary.



FDI — Fuel Danger Index

Data Sources: MODIS MOD13Q1 (NDVI) + Landsat 9 SR (NDMI) **Output Resolution:** 30 m

Construction:

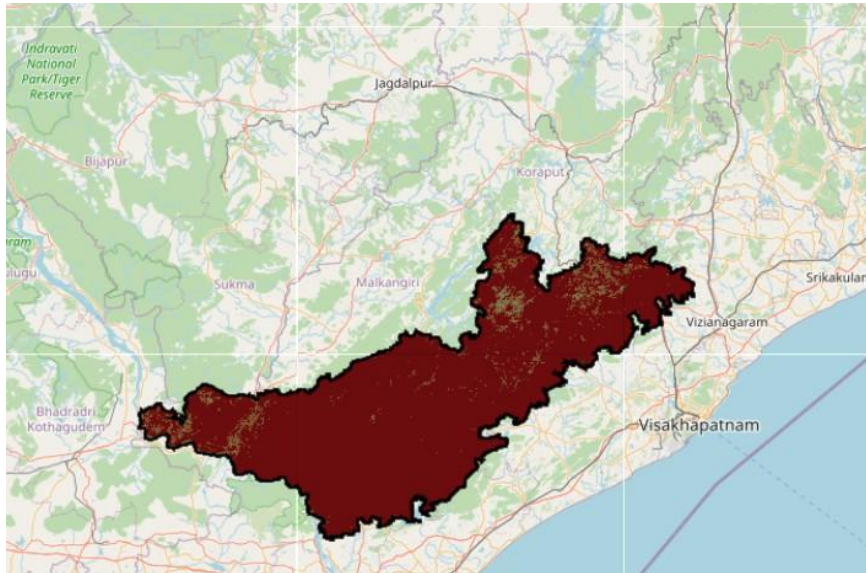
- FDI is computed as the sum of NDVI and NDMI layers, both resampled to 30 m.
- $$\text{FDI} = \text{NDVI}_{30\text{m}} + \text{NDMI}_{30\text{m}}$$

Role in Fire Analysis:

- Integrates fuel quantity (NDVI) and fuel dryness (NDMI) into a single layer.
- High FDI areas have dense, dry vegetation — greatest fire spread potential.
- FDI feeds directly into the AHP composite wildfire risk model as the fuel component.

Processing:

- Both layers normalized to 0–1 before summation to ensure equal weighting.
- Output clipped to ASR district and exported at 30 m resolution.



ADI — Activity Danger Index

Data Sources: Oxford MAP Accessibility Dataset (GEE) + GPW v4 / WorldPop **Output**

Resolution: 100 m

Components:

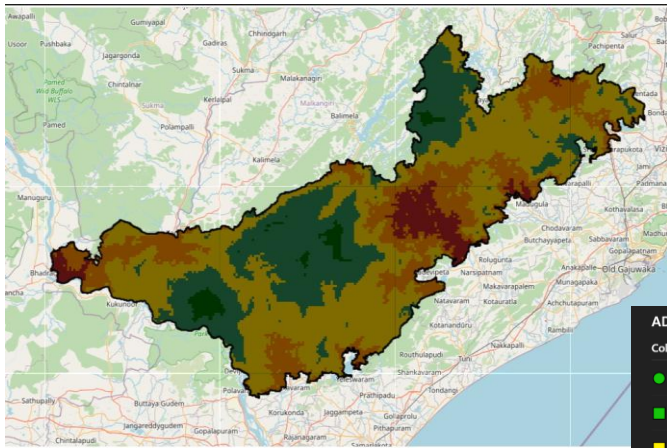
- **ADI_r (Accessibility):** Travel-time to nearest city — shorter travel time indicates higher human presence and access.
- **ADI_d (Population Density):** Population per pixel as a proxy for anthropogenic ignition pressure.

Role in Fire Analysis:

- Captures human-caused fire risk from activities such as shifting cultivation, grazing and road use.
- Areas near settlements and road networks score highest on ADI.

Processing:

- Both components normalized to 0–1 and combined into a single ADI layer.
- Inverted accessibility raster used so proximity = higher risk.



ADI Map Legend

Color	ADI Class	Meaning	
Dark Green	1	Very Low Human Disturbance	
Light Green	2	Low Disturbance	
Yellow	3	Moderate Disturbance	
Orange	4	High Disturbance	
Red	5	Very High Human Disturbance	

WDI — Weather Danger Index

Data Source: ERA5-Land Monthly Aggregates (ECMWF via GEE)
~9 km (resampled to 100 m)

Spatial Resolution:

Study Period: February – April 2023 (confirmed fire season)

Input Variables:

- 2 m air temperature
- Relative humidity (derived from dewpoint temperature)
- Wind speed (10 m U and V components)

Role in Fire Analysis:

- High temperature, low humidity and strong winds collectively increase fire ignition probability and spread rate.
- WDI normalizes and combines all three variables into a single weather risk score.

Processing:

- Monthly mean values computed over the fire-season window and normalized to 0–1.

TDI — Topographic Danger Index

Data Source: USGS SRTM DEM **Spatial Resolution:** 30 m

Input Variables:

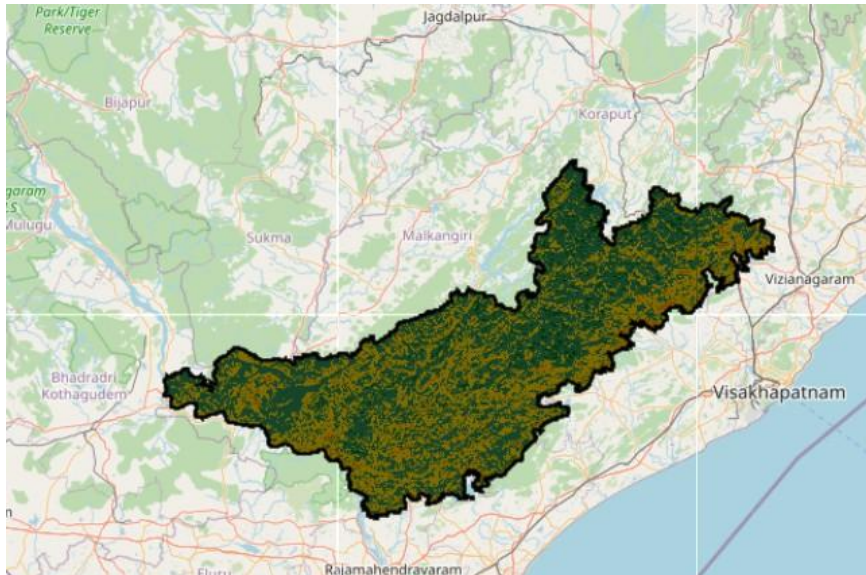
- **Slope:** Derived from DEM; steeper terrain accelerates fire spread uphill.
- **Aspect:** South- and southwest-facing slopes receive more solar radiation, remain drier and are more fire-prone.

Role in Fire Analysis:

- Terrain controls the direction and speed of fire movement independent of fuel or weather.
- TDI identifies topographically vulnerable areas even under moderate fire weather.

Processing:

- Slope and aspect layers each normalized to 0–1.
- Combined into TDI as a weighted sum; clipped to ASR district boundary.



dNBR — Burn Severity

Data Source: Landsat 9 Surface Reflectance (B5) and SWIR-2 (B7)

Spatial Resolution: 30 m

Bands: NIR

Study Periods:

- Pre-fire NBR: November – December 2022
- Post-fire NBR: May – June 2023

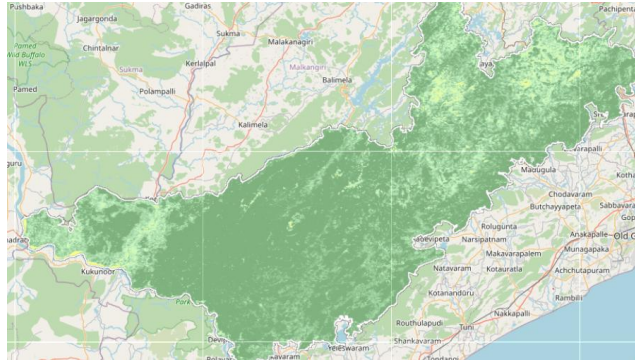
Role in Fire Analysis:

- NBR isolates burned areas by contrasting near-infrared reflectance (healthy vegetation) with shortwave-infrared reflectance (char/ash).
- $\Delta\text{NBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}$; higher values indicate greater burn severity.

Processing:

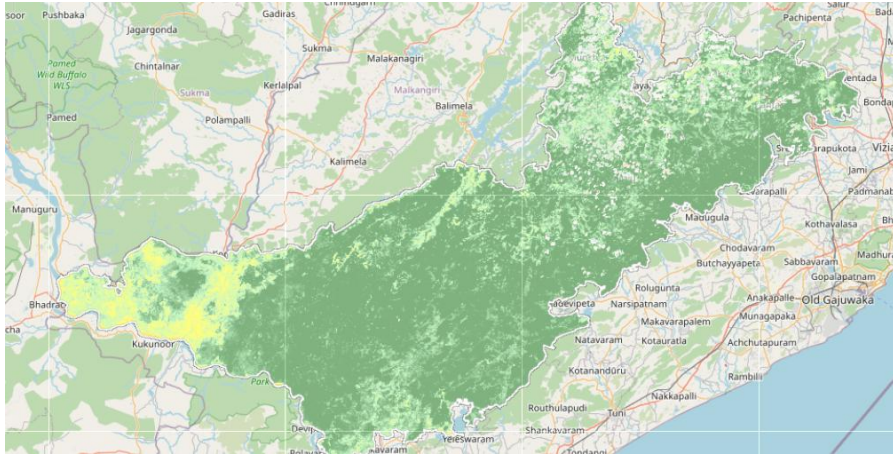
- Cloud masking applied using QA_PIXEL; images composited over each window.
- dNBR classified into standard severity classes: Unburned, Low, Moderate and High severity.

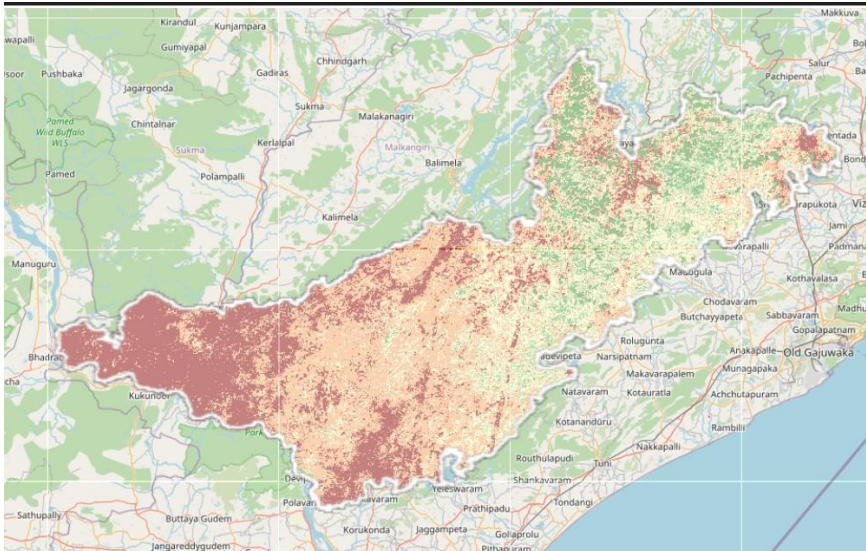
Pre Fire NBR



Color	Value Range	Interpretation
Red	-0.7 to -0.3	Bare soil / water / non-vegetated
Orange	-0.3 to 0.0	Sparse vegetation
Yellow	0.0 to 0.2	Moderate vegetation
Light Green	0.2 to 0.4	Healthy vegetation
Dark Green	0.4 to 0.6	Dense vegetation

Post fire NBR





AHP — Composite Wildfire Risk Model

Method: Analytic Hierarchy Process (AHP) Weighted Linear Combination **Output**

Resolution: 100 m

Input Factors (8): LST, Elevation, NDVI, Moisture Index, LULC, Aspect, Wind Speed, Slope

Weighting:

- Pairwise comparison matrix constructed based on relative fire-risk importance of each factor.
- Consistency Ratio **CR** < **0.1** confirms the matrix is logically consistent.

Composite Risk Formula:

$$\text{Risk} = \sum_{i=1}^8 w_i \cdot X_i$$

Output & Validation:

- Risk map classified into five zones: Very Low, Low, Moderate, High, Very High.
- Validated against MODIS FIRMS active fire detections for Feb–Apr 2023.

AHP Weight Verification:

Factor	Computed	Paper	Match

NDVI	0.1683	0.1645	⚠
Temperature	0.3373	0.3498	⚠
Slope	0.0217	0.0204	✅
Moisture Index	0.0767	0.0725	⚠
Elevation	0.2370	0.2456	⚠
Aspect	0.0444	0.0401	⚠
Wind Speed	0.0416	0.0378	⚠
LULC	0.0731	0.0693	⚠

Consistency Check:

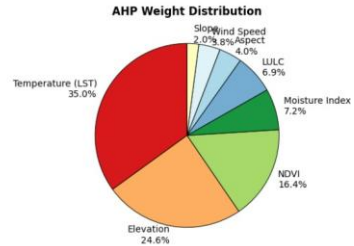
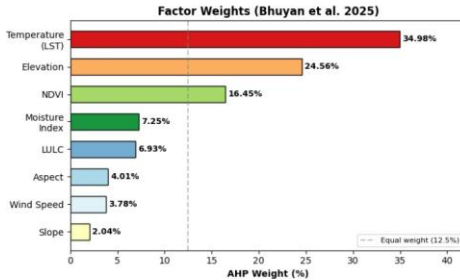
$\lambda_{\max}$ = 8.7699

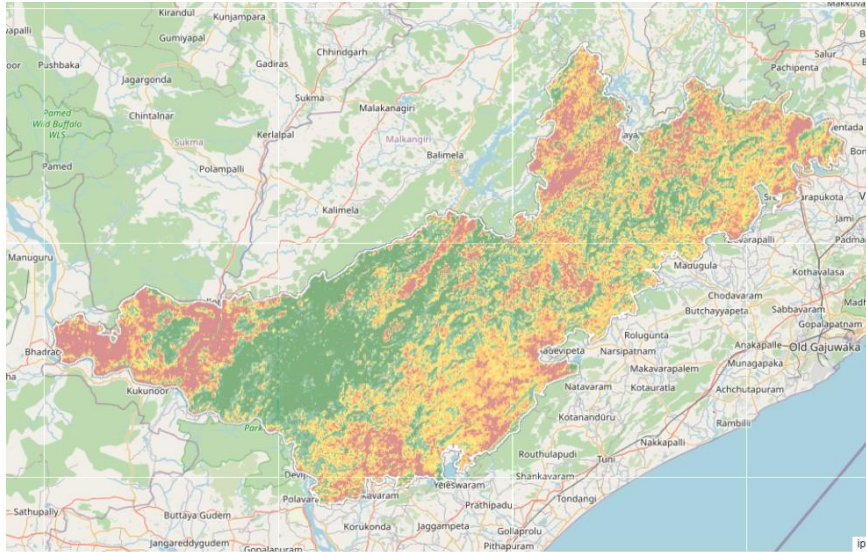
CI = 0.1100

CR = 0.0780 ✅ Consistent (CR < 0.1)

✅ Using published AHP weights from Bhuyan et al. (2025), Table 3

AHP Weights — Forest Fire Risk Factors
(Bhuyan et al. 2025, Table 3 | CR = 0.07)





LULC Area Statistics for ASR District:

ESA Class	Risk Class	Area km ²

Tree cover	5	8766.2
Shrubland	4	2.9
Grassland	4	2573.5
Cropland	3	1133.1
Built-up	2	13.3
Bare/sparse	1	18.4
Perm. water	1	125.1
Herb. wetland	2	3.6

✅ LULC classified – Forest (Tree cover) = Class 5 (highest fire risk)

AHP Forest Fire Risk Zone Distribution

AHP Forest Fire Risk Zone Distribution – Alluri Seetaramaraju:

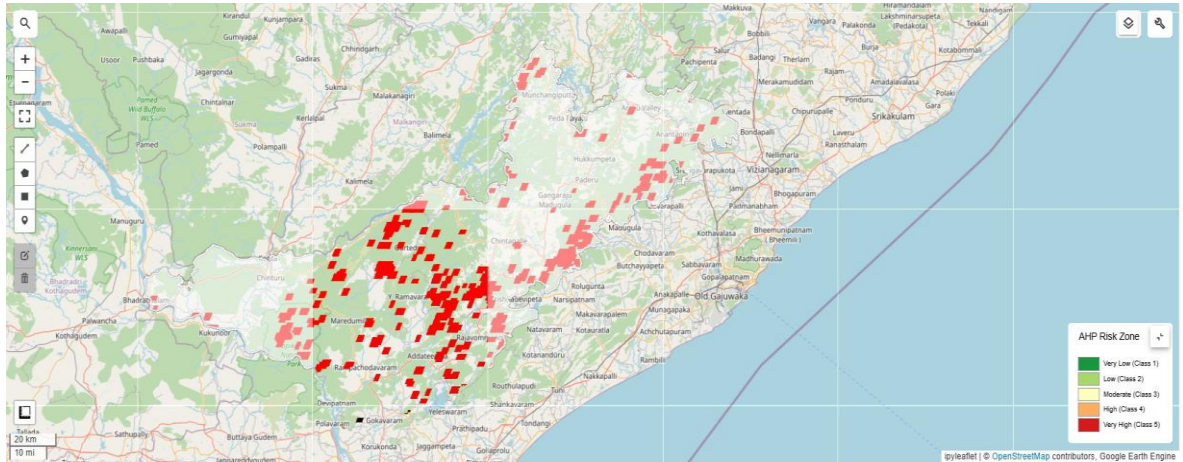
Zone	Area km ²	%
Very Low	2531.27	19.97
Low	2509.36	19.79
Moderate	2550.75	20.12
High	2509.03	19.79
Very High	2535.66	20.00

Total area accounted: 12636 km²

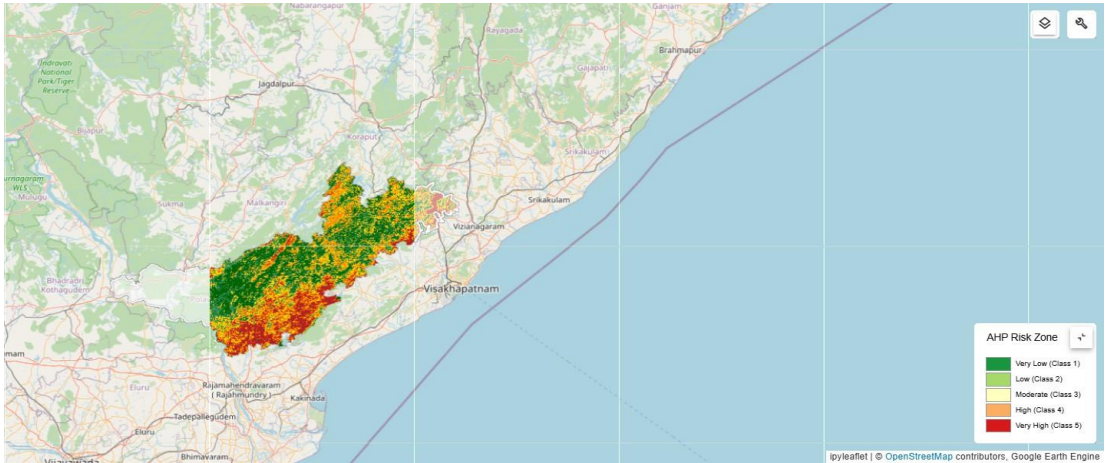
Paper results for Malkangiri (Table, Section 4.9):

Very Low	: 40.91%
Low	: 33.06%
Moderate	: 13.57%
High	: 12.45%
Very High	: 0.00%

FIRMS Fire Points



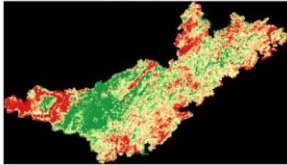
Validation with FIRMS Fire Points



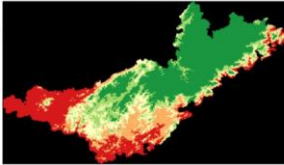
AHP Factor maps

AHP Factor Maps — Alluri Seetaramaraju District (Class 1=Low Risk, 5=High Risk)

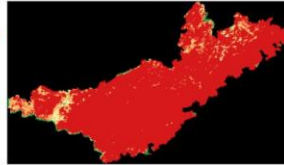
Temperature (LST)



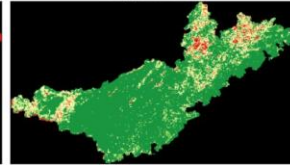
Elevation



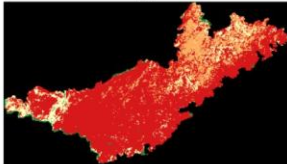
NDVI



NDMI (Moisture Index)



LULC



Aspect



Wind Speed



Slope



Class 1: Very Low Class 2: Low Class 3: Moderate Class 4: High Class 5: Very High

Temporal Analysis Overview

Purpose: Multi-year temporal analysis of fire activity and biophysical drivers over 2018–2024, extending the single-year (2022–2023) index analysis.

Scientific Questions:

- | | |
|---|---------------------------------|
| • Q1: Is fire activity increasing over 2018–2024? | <i>Mann-Kendall trend test</i> |
| • Q2: Which years were anomalously severe? | <i>Anomaly vs. 7-year mean</i> |
| • Q3: What drives bad fire years? | <i>Pearson correlation</i> |
| • Q4: Is vegetation recovering between fire years? | <i>Seasonal NDVI recovery</i> |
| • Q5: Are the same areas burning repeatedly? | <i>Fire return interval map</i> |

Baseline: 7-year mean (2018–2024). All anomalies are expressed as deviations from this mean, following standard fire ecology practice.

A1 — FIRMS Annual Fire Pixel Counts (2018–2024)

Data Source: MODIS FIRMS **Band:** T21 **Resolution:** 1 km **Threshold:** $T_{21} > 300\text{ K}$

Method:

- Monthly fire pixel counts queried for all 84 months (2018–2024).
- Annual totals computed by summing monthly counts per year.
- 7-year mean baseline and standard deviation computed:

$$\bar{F} = \frac{1}{7} \sum_{y=2018}^{2024} F_y, \quad z_y = \frac{F_y - \bar{F}}{\sigma_F}$$

Results (from notebook output):

Year	Fire Pixels	z-score	Rank	Status
2018	6,914	+0.54	3	Above mean
2019	7,817	+1.04	2	Severe
2020	3,227	-1.47	7	Below mean
2021	7,942	+1.11	1	Severe
2022	4,378	-0.84	6	Below mean
2023	4,713	-0.66	5	Below mean

Six indices computed per year using GEE Python API over the confirmed fire season (Feb–Apr):

Spectral Indices:

- **NDVI** — MODIS MOD13Q1, annual median, 250 m; scaled $\times 0.0001$
- **NDMI** — Landsat 8/9, $\frac{B5-B6}{B5+B6}$, 30 m
- **FDI** — $NDVI_{30m} + NDMI$, 30 m
- **dNBR** — $NBR_{pre} - NBR_{post}$; pre = Dec(y-1) + Jan(y); post = Mar–Apr(y)

Landsat sensor continuity: L8 used for 2018–2021; L9 for 2022–2024. Identical band designations (B5/B6/B7) ensure cross-sensor consistency.

Meteorological Variables:

- **LST** — Landsat ST_B10, converted to °C:
 $T_{LST} = 0.00341802 \times DN + 149.0 - 273.15$
- **WDI** — ERA5-Land fire-season means: temperature, relative humidity (Magnus formula), wind speed ($\sqrt{u^2 + v^2}$)

A2 — Output: Annual Index Values (2018–2024)

Year	NDVI	LST (°C)	dNBR	WDI	FDI	Fire Pixels
2018	0.574	36.958	0.231	1.994	0.509	6914.478
2019	0.576	37.609	0.218	1.898	0.521	7817.063
2020	0.622	36.163	0.218	1.697	0.592	3226.537
2021	0.630	34.675	0.227	1.918	0.581	7941.553
2022	0.615	34.664	0.259	1.917	0.625	4378.498
2023	0.637	35.584	0.158	1.934	0.622	4713.059
2024	0.617	36.945	0.231	2.058	0.584	6453.435

Annual mean values of NDVI, LST, dNBR, WDI, FDI and detected fire pixels computed using GEE for 2018–2024.

Q1 — Trend Analysis: Mann-Kendall Test (2018–2024)

Method: Non-parametric Mann-Kendall test — no normality assumption; suited for $n = 7$ environmental time series.

Test statistic:

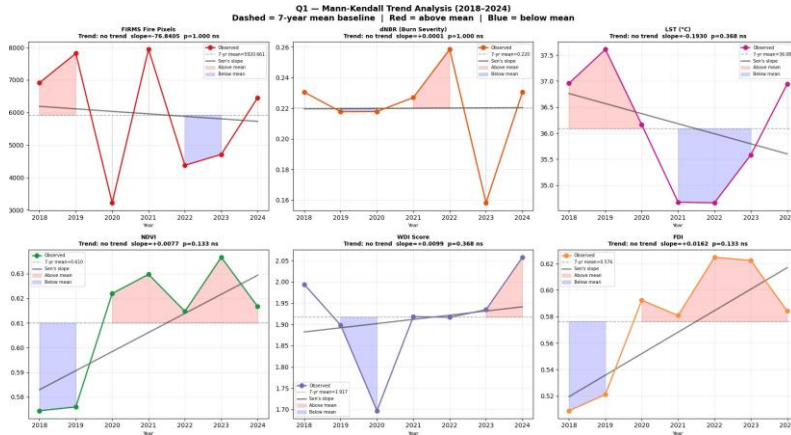
$$S = \sum_{k < j} \text{sgn}(x_j - x_k), \quad \hat{\theta} = \text{median} \left(\frac{x_j - x_k}{j - k} \right) \quad (\text{Sen's slope})$$

Variables tested: FIRMS fire pixels, dNBR, LST, NDVI, WDI, FDI

Result (FIRMS): No significant trend detected
(Sen slope = -76.84 px/yr, $p = 1.000$, not significant)

Visualisation: 2×3 subplot grid; each panel shows the observed series, 7-year mean baseline (dashed), Sen's slope trend line, and red/blue shading for above/below-mean years.

Q1 — Mann-Kendall Trend Analysis (2018–2024)



Each panel shows annual values (2018–2024) with the 7-year mean baseline (dashed) and Sen's slope trend line.

Q2 — Anomaly Analysis: Which Years Were Severe?

Method: Each index anomaly normalized to $[-1, +1]$:

$$a_c^* = \frac{x_c - \bar{x}_c}{\max |x_c - \bar{x}_c|}$$

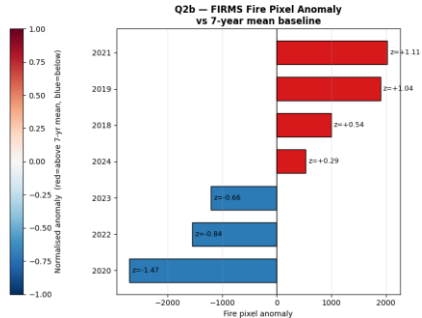
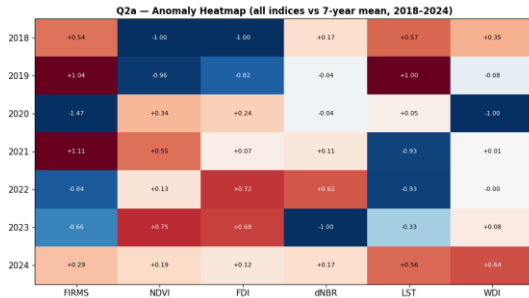
Output Q2a — Anomaly heatmap: Rows = years, columns = FIRMS, NDVI, FDI, dNBR, LST, WDI. Diverging RdBu_r colormap; red = above 7-year mean; blue = below.

Output Q2b — FIRMS anomaly bar chart: Years ranked by fire pixel anomaly; each bar annotated with its z-score.

Key finding:

- **2021** was the worst fire year ($z = +1.11$, rank 1 of 7).
- **2019** was the second most severe ($z = +1.04$, rank 2).
- **2020** had the lowest fire activity ($z = -1.47$), coinciding with COVID-related mobility restrictions.

Q2 — Anomaly Patterns Across Years (2018–2024)



- Heatmap shows normalized anomalies of fire activity and environmental indices relative to the 7-year mean.
- Red indicates above-average conditions; blue indicates below-average conditions.
- FIRMS anomaly bar chart ranks years by fire activity deviation.

Q3 — Driver Correlation Analysis

Method: Pearson correlation coefficient between annual FIRMS fire pixels and each candidate driver:

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \cdot \sum_i (Y_i - \bar{Y})^2}}$$

Two-tailed p -value computed via `scipy pearsonr`; NaN values excluded pairwise.

Drivers tested (9): LST, ERA5 air temperature, relative humidity, wind speed, NDVI, NDMI, FDI, WDI, dNBR

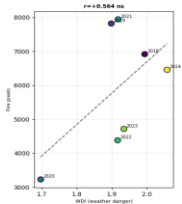
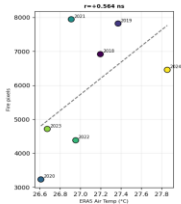
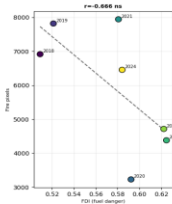
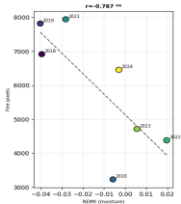
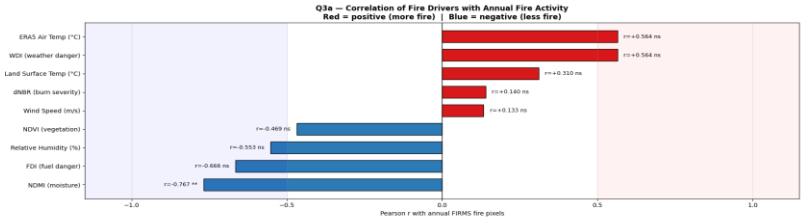
Key finding:

- **Top driver: NDMI (moisture)** with $r = -0.767$
- Negative correlation: lower canopy moisture $\rightarrow$ more fire activity.
- This confirms that fuel dryness is the primary modulator of fire intensity in ASR district.

Visualisation: Sorted horizontal bar chart ($|r|$ order) + scatter plots for top-4 drivers with year-coloured markers (Viridis) and OLS trend lines.

Q3 — Fire Driver Correlation Results (2018–2024)

Q3 — What Drives Bad Fire Years? (2018–2024)



Q3 — Key Findings from Driver Correlation

Interpretation:

Pearson correlation analysis was used to examine relationships between annual fire pixels and environmental drivers.

NDMI (vegetation moisture) shows the strongest correlation with fire activity ($r = -0.767$).

The negative relationship indicates that **lower canopy moisture corresponds to higher fire activity**.

Q4 — Post-Fire Vegetation Recovery (NDVI)

Data Source: MODIS MOD13Q1, 250 m, median reducer

Three seasonal windows per year:

- **Pre-fire** (January): baseline greenness
- **Fire season** (Feb–Apr): minimum greenness under fire stress
- **Post-monsoon** (Aug–Oct): recovery assessment

Recovery Rate Formula:

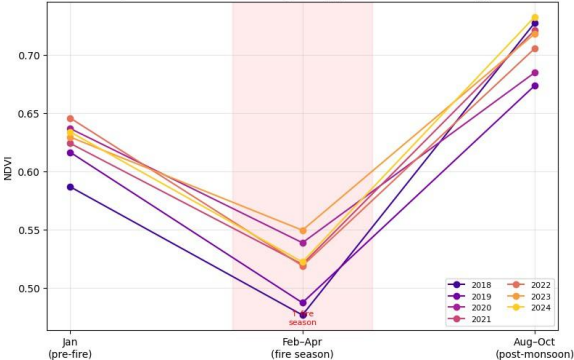
$$R_y = \frac{\text{NDVI}_{\text{post}} - \text{NDVI}_{\text{peak}}}{|\text{NDVI}_{\text{pre}} - \text{NDVI}_{\text{peak}}|} \times 100\%$$

Key finding: 2019 had the weakest recovery ($R = 144.2\%$), despite being the second highest fire year, suggesting post-fire biomass accumulation was suppressed. All years exceeded 100% recovery, indicating strong monsoon-driven vegetation rebound in ASR district.

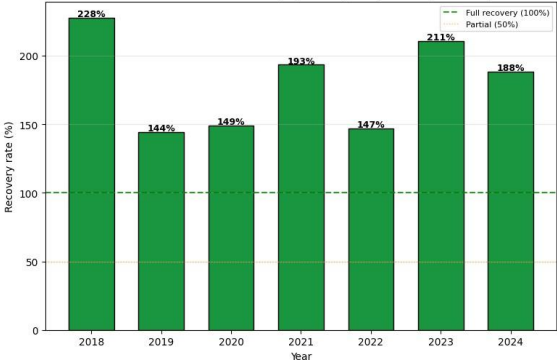
Q4 — Post-Fire Vegetation Recovery (NDVI)

Q4 — Vegetation Recovery Analysis — ASR District (2018-2024)

Q4a — Seasonal NDVI Trajectory (pre → fire → recovery)



Q4b — Post-Monsoon NDVI Recovery Rate (%)
Green ≥ 100% = full recovery | Orange = partial | Red = poor



Q5 — Fire Return Interval Map

Method: For each year, a binary fire detection layer is derived from FIRMS ($T_{21} > 300\text{ K}$). The 7 annual layers are stacked and summed pixel-wise:

$$\text{FireCount}_{xy} = \sum_{y=2018}^{2024} \mathbf{1}[\text{fire detected at } (x, y) \text{ in year } y]$$

Result: integer map $[0, 7]$ at 1 km resolution.

Area quantification:

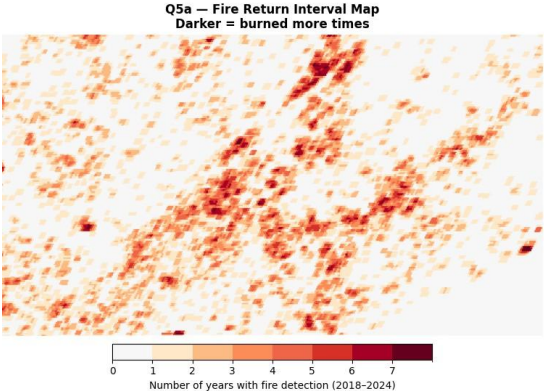
$$A_n = \sum_{xy} \mathbf{1}[\text{FireCount}_{xy} = n] \times \text{pixelArea} \div 10^6 \text{ (km}^2\text{)}$$

Key finding:

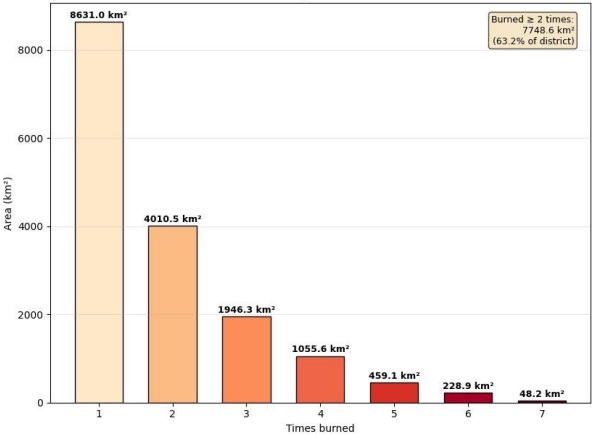
- **7,748.6 km²** (63.2% of ASR district) burned at least twice in 2018–2024.
- Confirms widespread **repeat burning** — a major obstacle to long-term forest recovery.

Q5 — Fire Return Interval Map

Q5 — Fire Return Interval — ASR District (2018-2024)



Q5b — Area Burned by Return Frequency
(km² burned exactly N times in 2018-2024)



Temporal Summary: Key Findings (2018–2024)

Year	Status	Rank	LST (°C)	NDVI	dNBR	Recovery %
2018	Above mean	3	36.96	0.574	0.231	227.8
2019	Severe	2	37.61	0.576	0.218	144.2
2020	Below mean	7	36.16	0.622	0.218	148.9
2021	Severe	1	34.68	0.630	0.227	193.4
2022	Below mean	6	34.66	0.615	0.259	147.0
2023	Below mean	5	35.58	0.637	0.158	210.8
2024	Above mean	4	36.95	0.617	0.231	188.4

- **Q1:** No significant fire trend ($p = 1.000$; Sen slope = -76.84 px/yr)
- **Q2:** Worst year — **2021** ($z = +1.11$)
- **Q3:** Top driver — **NDMI (moisture)** ($r = -0.767$)
- **Q4:** Worst recovery — **2019** (144.2%)
- **Q5:** Repeat burning — **7,748.6 km² (≥ 2 fires)** = 63.2% of district

Biodiversity Dataset Sources

Species records obtained from GBIF and published Eastern Ghats biodiversity studies.

- Each record contained latitude–longitude coordinates of species observations.
- Coordinates were converted into spatial points using GIS/Python tools.
- Data was clipped using the Alluri Sitaramaraju district boundary shapefile.
- Only records within the district and with valid georeferenced coordinates were retained.
- Information was cross-checked with FSI (ISFR reports) and ecological studies.

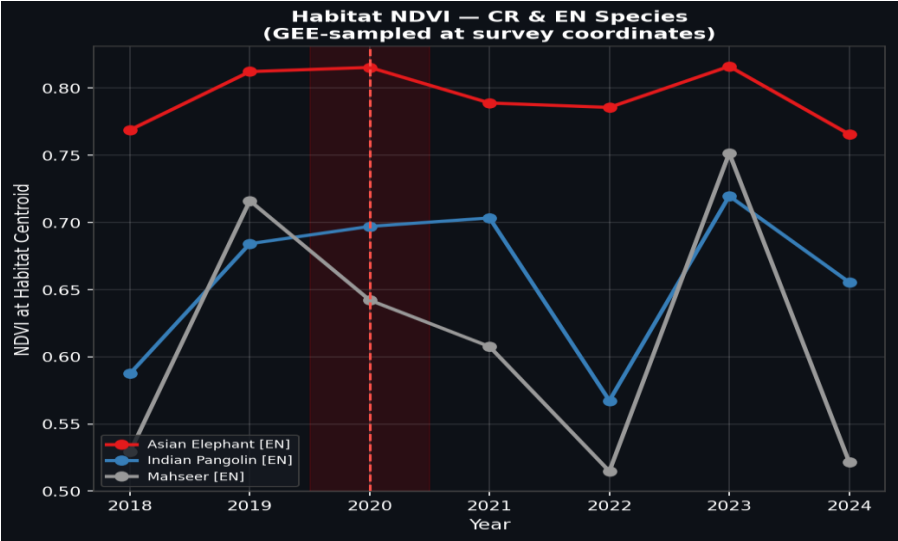
Outcome:

Clean spatial dataset of species occurrence points within the ASR district used for analysis.

Species Taxonomic Categories

Category	Description	Example Species
Mammals	Warm-blooded vertebrates with hair/fur and live birth	Asian Elephant, Sloth Bear
Birds	Feathered vertebrates capable of flight (most species)	Great Hornbill, Alexandrine Parakeet
Reptiles	Cold-blooded vertebrates with scales	King Cobra, Indian Rock Python
Others	Includes fish, amphibians, plants, etc.	Mahseer, Purple Frog, Bedome's Cycad

Habitat NDVI (2018-2024)



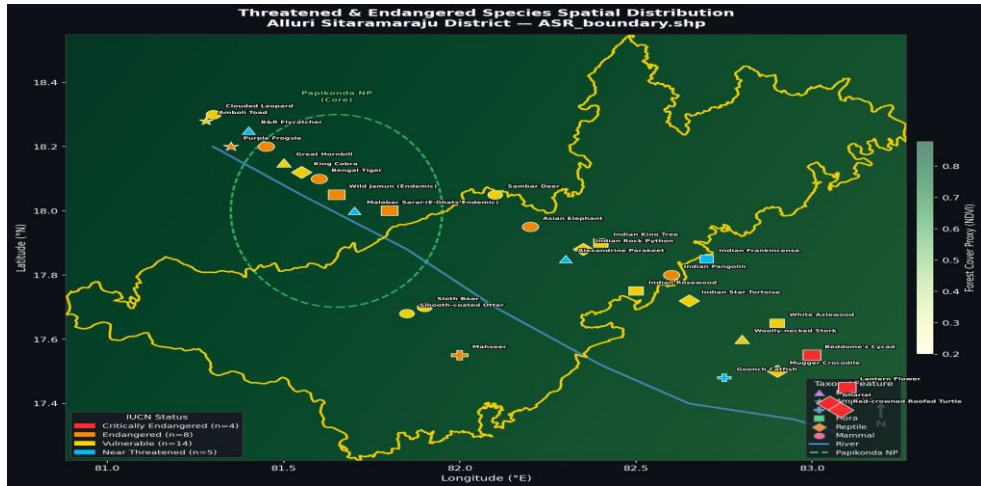
Habitat NDVI Trends (2018–2024)

- Asian Elephant:** NDVI remains consistently high, indicating **stable and healthy vegetation cover** in its habitat.
- Indian Pangolin:** Shows **moderate fluctuations**, suggesting temporary vegetation stress but overall recovery after disturbance.
- Mahseer:** Displays **larger NDVI variability**, indicating higher sensitivity of its habitat to environmental changes.

Key Insight:

The highlighted **2020 fire period** corresponds to a short-term drop in vegetation health, followed by **gradual ecosystem recovery in subsequent years**.

Species Distribution Map



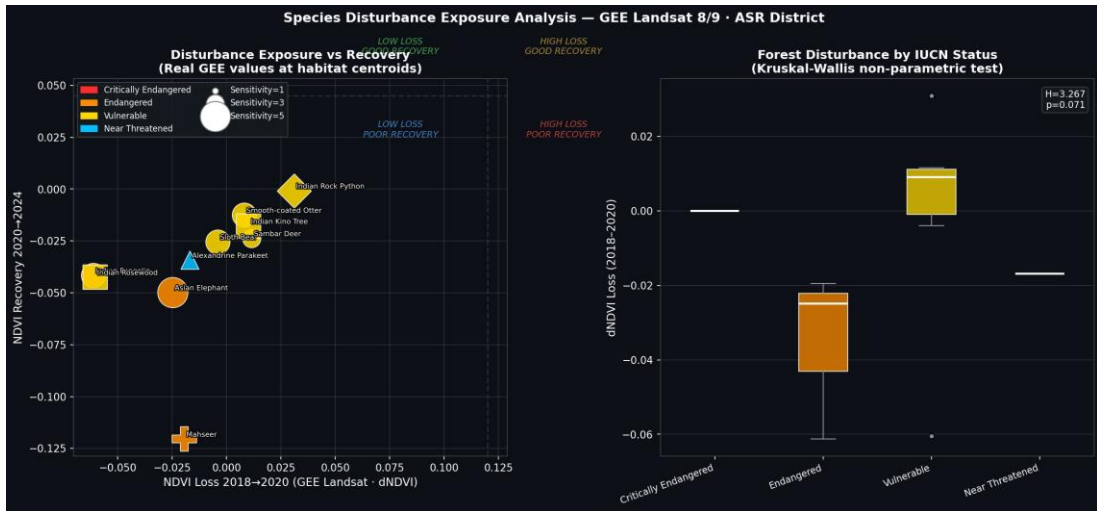
Species Distribution Analysis

- Species observations are clustered mainly in the Papikonda NP region, indicating a key biodiversity hotspot.
- The forest cover gradient (background shading) shows most species occur in moderate to dense forest areas, highlighting their dependence on intact habitats.
- Several species are distributed along river corridors, suggesting the importance of riparian ecosystems.

IMPORTANT:

Papikonda National Park and surrounding forest landscapes act as a **critical refuge for threatened and endangered species in the district.**

Habitat Disturbance Analysis



IUCN Conservation Status Categories

IUCN Category	Meaning	Conservation Risk
CR — Critically Endangered	Extremely high risk of extinction	Very High
EN — Endangered	Very high risk of extinction in the wild	High
VU — Vulnerable	High risk of becoming endangered	Moderate
NT — Near Threatened	Close to qualifying as threatened	Lower

Disturbance vs Recovery Analysis (Left Plot)

- The x-axis represents vegetation loss (dNDVI) during the 2018–2020 fire period, while the y-axis represents vegetation recovery from 2020–2024.
- Species positioned toward the left experienced lower vegetation loss, whereas those further right experienced greater disturbance.
- Species located lower on the plot show weaker recovery, indicating higher habitat vulnerability after the disturbance.
- Bubble size represents species sensitivity, where larger bubbles indicate species that are more vulnerable to habitat disturbance.

Key insight:

Species that experienced higher vegetation loss with weaker recovery are more exposed to disturbance and may require stronger conservation attention.

Disturbance comparison by IUCN status (Right Plot)

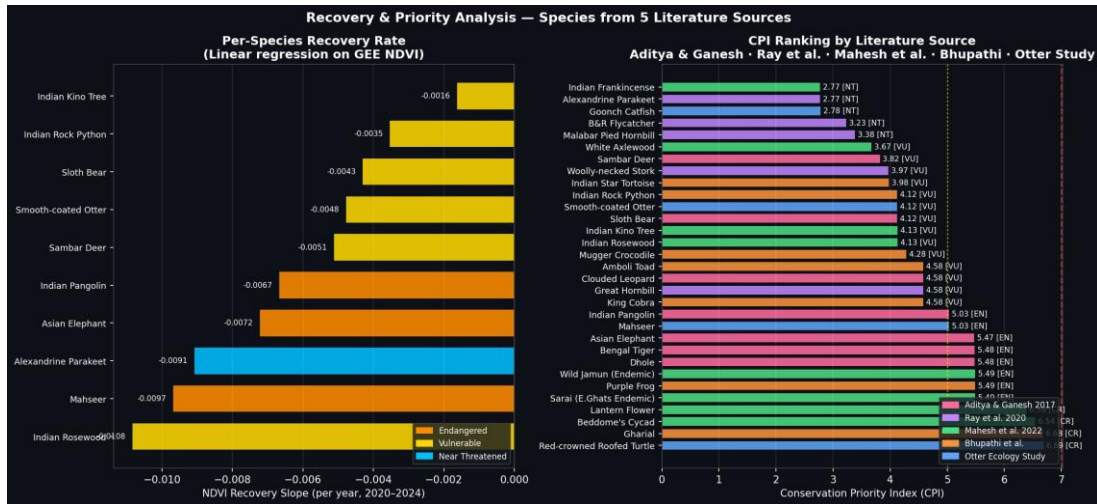
The boxplot compares vegetation loss (NDVI decrease) during the fire period (2018–2020) across different IUCN conservation categories.

- Endangered species habitats show relatively higher vegetation loss, while Vulnerable and Near Threatened categories show mixed disturbance patterns.
- The spread of the boxplots indicates variability in disturbance exposure within each conservation category, reflecting spatial heterogeneity.
- The Kruskal–Wallis test result ($H \approx 3.27$, $p \approx 0.07$) suggests moderate variation, but the differences are not statistically significant.

Key insight:

The result is not strong because disturbance values are similar across categories and overlap, so the difference is not clearly distinguishable.

Forest Recovery Priority Areas



Species Habitat Recovery Analysis (Left Plot)

The horizontal bar plot shows NDVI recovery trends (2020–2024) for species habitats using Landsat NDVI time-series from Google Earth Engine.

- Recovery was estimated using linear regression on NDVI values, producing a yearly recovery slope.
- Species with slopes closer to 0 indicate better vegetation recovery, while more negative slopes indicate slower ecosystem recovery.
- Habitats of Mahseer and Indian Rosewood show the weakest recovery, suggesting potential ecological stress in those areas.

Key insight:

Species habitats differ in recovery rates, indicating variation in ecosystem resilience across the district.

Conservation Priority Index (CPI) Ranking

- The CPI was calculated to prioritize species for conservation based on multiple ecological factors.
- Each species was assigned a score using IUCN conservation status, ecological importance, and presence in multiple literature sources.
- Higher weights were given to more threatened categories (CR > EN > VU > NT).
- These scores were combined to generate a Conservation Priority Index (CPI) for each species.
- Species were then ranked by CPI to identify those requiring the most urgent conservation attention.

Key insight:

Higher CPI values indicate species with higher extinction risk and greater conservation priority.

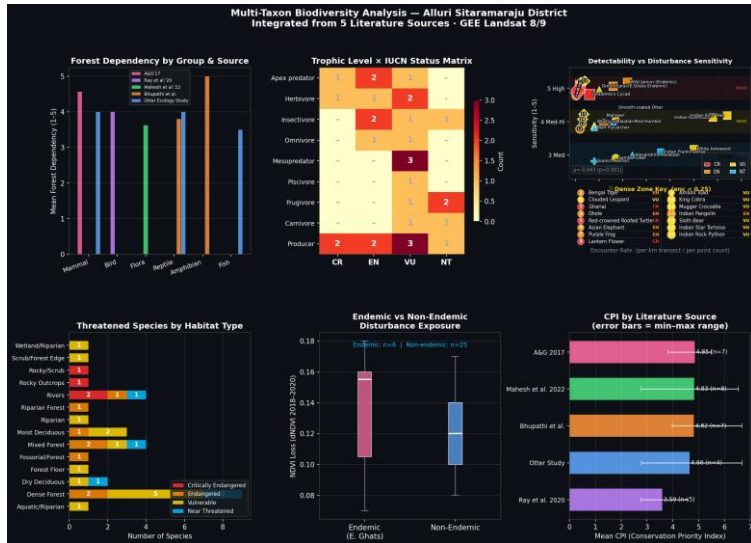
Conservation Priority Index Ranking (Right Plot)

- The bar chart ranks species using a Conservation Priority Index (CPI) derived from five literature sources.
- CPI integrates IUCN conservation status, ecological importance, and literature occurrence.
- Species with higher CPI values represent higher conservation priority.
- The ranking highlights species such as Red-crowned Roofed Turtle, Gharial, Bengal Tiger, and Dhole as requiring urgent conservation attention.

Key insight:

CPI helps prioritize species for conservation planning by identifying those with the greatest ecological and extinction risk.

Biodiversity Impact Dashboard



- Integrates disturbance and recovery analysis across species habitats.
- Highlights regions where biodiversity has been significantly affected.
- Identifies habitats most vulnerable to forest disturbances.
- Shows locations where vegetation recovery is strong.
- Supports biodiversity monitoring and conservation planning.

Key Findings from Species Analysis

- NDVI analysis reveals a noticeable vegetation decline around **2020**.
- Disturbances likely caused by forest fires and human activities such as shifting cultivation.
- Several areas show gradual vegetation recovery after **2021**.
- Some habitats still exhibit slow recovery, indicating ecological stress.
- Results highlight priority regions for biodiversity conservation and forest management.

Conclusion

- Multi-temporal satellite analysis helps monitor forest changes
- Remote sensing techniques effectively detect disturbances
- Results can support sustainable forest management
- Temporal analysis (2018–2024) shows **no statistically significant fire trend**, though **2021** stands out as the most severe year ($z = +1.11$)
- **Fuel moisture (NDMI)** is the strongest predictor of annual fire intensity ($r = -0.767$)
- Over **63%** of the district has burned at least twice, highlighting the need for targeted firebreak and restoration strategies

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